

## 1. Crisis Timeline (November 2025 – April 2026)

| DATE          | EVENT                                      | SIGNIFICANCE                                                                                                        |
|---------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Late Nov 2025 | Cereulide detected at Nestlé Nunspeet (NL) | Routine testing following new equipment commissioning flagged toxin in infant formula; origin not yet traced.       |
| 10 Dec 2025   | "Silent withdrawal"                        | Nestlé informs Dutch NVWA and European Commission; 25 batches withdrawn across 16 countries without public notice.  |
| 24 Dec 2025   | Source traced to ARA oil                   | Contamination confirmed in crude ARA oil supplied by a Chinese fermentation firm.                                   |
| 29 Dec 2025   | Supplier notified                          | CABIO Biotech informed; traceback indicates contamination extends to batches produced as early as 2024.             |
| 5 Jan 2026    | Public recall launched                     | Nestlé brands SMA, Beba, Guigoz and Alfamino pulled across 60+ countries.                                           |
| Jan 2026      | Danone joins                               | Aptamil, Nutrilon recalls across Europe, UK and Asia.                                                               |
| 22 Jan 2026   | Lactalis recalls                           | Fourth major manufacturer; Hochdorf and Vitagermine follow.                                                         |
| 29 Jan 2026   | Recall extended                            | Nestlé expands to additional batches; experts call for EU cereulide safety limits.                                  |
| 2 Feb 2026    | EFSA scientific opinion                    | Derived limits: 0.054 µg/L (infant formula), 0.1 µg/L (follow-on); ARfD 0.014 µg/kg bw.                             |
| 13 Feb 2026   | 7 countries investigating cases            | Gastrointestinal symptoms in infants post-consumption reported.                                                     |
| 16 Feb 2026   | French preliminary investigations opened   | <i>Enquêtes préliminaires</i> against Nestlé, Danone and Lactalis; two infant deaths formally linked to the probes. |
| 24 Feb 2026   | EU Implementing Regulation                 | Emergency import controls on Chinese ARA oil — published under urgent procedure.                                    |
| 25 Feb 2026   | WHO DON596 published                       | 99 countries confirmed; 144 suspected/confirmed cases; moderate public health risk.                                 |
| 26 Feb 2026   | EU controls effective                      | 100% documentary + 50% physical/identity checks at BCPs; mandatory Chinese cereulide-absence certificate.           |
| Mar 2026      | Third infant death reported                | Third French infant death under preliminary review; CABIO Biotech share price falls sharply.                        |
| Apr 2026      | Industry-wide reassessment                 | GRAS notifications under review; supply-chain restructuring and supplier-qualification programmes underway.         |

**Response-delay metrics:** 26 days from first internal detection (Nestlé Nunspeet, late Nov 2025) to public recall (5 Jan 2026); 12 days from confirmed root-cause (24 Dec 2025) to public recall.

## 2. Ingredient Overview & Regulatory Status

| ATTRIBUTE           | DETAIL                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical identity   | Arachidonic acid, C20:4 n-6 — long-chain polyunsaturated omega-6 fatty acid.                                                                                                                                                                                                                                                                                                       |
| Production organism | <i>Mortierella alpina</i> (oleaginous filamentous fungus; not GMO; not pathogenic).                                                                                                                                                                                                                                                                                                |
| Process             | Submerged fermentation → biomass harvesting → drying → solvent or non-solvent extraction → degumming → bleaching → deodorisation → standardisation to ≥40% ARA.                                                                                                                                                                                                                    |
| Primary application | Infant formula (40–60 mg per 100 kcal; typically blended with DHA at 1:1 or 2:1 ARA:DHA ratio).                                                                                                                                                                                                                                                                                    |
| Other uses          | Adult nutritional supplements; clinical nutrition for premature infants; fortified dairy; sports nutrition.                                                                                                                                                                                                                                                                        |
| FDA status          | GRAS — Notices GRN 958 and GRN 1186 for fungal oil ≥40% ARA from <i>M. alpina</i> .                                                                                                                                                                                                                                                                                                |
| EU status           | Authorised under Commission Delegated Regulation (EU) 2016/127 for infant/follow-on formula. Addition is <b>optional</b> , capped at 1% of total fat content if added. The post-crisis policy debate is active: regulators face a fork between (a) making ARA mandatory with stringent sourcing controls, or (b) revisiting its functional necessity in the regulated composition. |
| Global acceptance   | Approved in 99+ countries for infant nutrition; Codex Alimentarius CXS 72-1981 framework.                                                                                                                                                                                                                                                                                          |

## 3. Supply Chain Structure — Manufacturers & Geography

### Leading Manufacturers

| COMPANY                        | HQ / KEY SITES                                                  | EST. SHARE      | NOTES                                                                                                                                                                                                                                                                          |
|--------------------------------|-----------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CABIO Biotech (Wuhan)          | Wuhan, Hubei, CN                                                | 35–40%          | Largest single global ARA oil producer. Originated as Cargill–Alking JV (2004); Cargill's North American ARA offering has historically been sourced through this relationship. Named by AFP as the source of 2025–26 cereulide contamination. Shares fell sharply post-recall. |
| DSM-Firmenich                  | Kaiseraugst (CH); Maastricht (NL); Kingstree SC (legacy Martek) | 25–30%          | Former Martek life'sARA™ line acquired by DSM in 2011. Kingstree SC is the principal non-Chinese fermentation site for infant-grade ARA. Schenectady, NY is a premix/blending facility (USD 10 M upgrade announced Mar 2026), not an ARA fermentation site.                    |
| Kingdomway                     | Multiple sites, CN                                              | 5–8%            | Chinese LCPUFA and fermentation specialist.                                                                                                                                                                                                                                    |
| Guangdong Runke Bioengineering | Guangdong, CN                                                   | 5–8%            | Expanding capacity aggressively in 2025–26.                                                                                                                                                                                                                                    |
| Suntory                        | Japan                                                           | <5%             | Specialised applications.                                                                                                                                                                                                                                                      |
| Cayman Chemical                | Ann Arbor MI, US                                                | <5%             | Research and speciality grades.                                                                                                                                                                                                                                                |
| Other Chinese specialists      | Wuhan Fuxing, Changsha Jiage,                                   | collective <10% | Smaller Chinese fermentation firms; infant-grade qualification status uncertain.                                                                                                                                                                                               |

| COMPANY | HQ / KEY SITES                                                                                                                                | EST. SHARE | NOTES |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
|         | Hubei Hengshuo,<br>Wuhan Weishunda,<br>Xuchang Yuanhua<br>(per market-intelligence sources; individual capacities not independently verified) |            |       |

The top 3–4 manufacturers collectively control ~75–90% of global supply. Cargill is not listed as a separate share-holder because its ARA market access is via the CABIO (ex-Cargill–Alking) joint venture and listing it independently would double-count.

## Geographic Concentration

| REGION              | ROLE                                                                         | KEY SITES                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Asia-Pacific</b> | Primary fermentation hub (~60–65% global capacity; China ~55–60%, Japan <5%) | China: CABIO (Wuhan), Guangdong Runke, Kingdomway, regional specialists. Japan: Suntory.                                                           |
| Europe              | Refining, formulation, blending                                              | DSM-Firmenich Maastricht (NL), Kaiseraugst (CH).                                                                                                   |
| North America       | Legacy fermentation + premix                                                 | DSM-Firmenich Kingstree SC (legacy Martek fermentation); Schenectady NY (premix blending, USD 10 M upgrade Mar 2026). Cayman Chemical (specialty). |
| Rest of World       | Limited                                                                      | Distribution and formulation only.                                                                                                                 |

## 4. Supply Chain Volumes & Market Projections

| METRIC                         | VALUE                                                       | SOURCE / NOTE                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global ARA market value (2025) | USD 248–275 million                                         | Mordor Intelligence, Valuates — speciality ARA oil only. Broader ARA chemical market reports cite up to USD 5.85 B when non-nutrition applications are included; scope difference, not a contradiction. |
| Projected market value (2031)  | USD 300–355 million                                         | CAGR 4.3–4.7% (nutrition segment).                                                                                                                                                                      |
| Production volume              | Several thousand MT per year                                | Format split: liquid oil ~60% / spray-dried microencapsulated powder ~40%.                                                                                                                              |
| Regional demand                | APAC 45–50% · Europe 25–30% · N. America 15–20% · RoW 5–10% | APAC projected highest CAGR to 2030.                                                                                                                                                                    |
| Typical price (infant grade)   | USD 50–100+/kg                                              | Premium reflects regulatory approvals + specialist fermentation barriers to entry.                                                                                                                      |

exposures; (v) consider acquisition or JV for algal/fermentation-based alternative lipid sources; (vi) prepare customer-facing technical dossier on post-crisis sourcing and cereulide control to accelerate qualification cycles.

## 11. Alternative Technologies & Diversification Pathways

| OPTION                                                        | HORIZON      | NOTES                                                                                                                         |
|---------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| Qualify additional <i>M. alpina</i> supplier (non-Chinese)    | 12–24 months | DSM-Firmenich Kingstree SC is the principal non-Chinese option. Infant-formula grade requires full regulatory approval cycle. |
| Non-solvent extraction (CABIO clean-label line)               | Existing     | Clean-label positioning; does not address cereulide risk at fermentation.                                                     |
| Algal ARA (engineered microalgae)                             | 2–4 years    | Earlier commercial stage than algal DHA; sustainability and traceability advantages.                                          |
| Synthetic biology / precision fermentation                    | 3–5 years    | Alternative hosts ( <i>Yarrowia lipolytica</i> , engineered yeasts); regulatory pathway for infant use unclear.               |
| Plant-based n-6 blends ( <i>Echium</i> , <i>Sacha inchi</i> ) | Existing     | Not identical to ARA; not authorised for infant formula; viable for adult/functional applications only.                       |
| Vertical integration / equity stake                           | 24–36 months | High capex; regulatory and technical barriers high.                                                                           |

## 12. Action Plan

### Key Takeaways

- Cereulide cannot be processed out of contaminated ARA oil — prevention at fermentation is the only control.
- Single-source dependency on Chinese fermentation is the primary supply-chain vulnerability.
- Fraud risk is elevated post-crisis.
- Legal exposure via French criminal investigations is an active, evolving risk; expected civil/consumer litigation is the larger downstream cost driver.

### Now (0–90 days)

| PRI. | ACTION                                                                                                                                                                    | TIMELINE  | OWNER                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------|
| P1   | Implement mandatory LC-MS/MS cereulide testing ( $\leq 0.054 \mu\text{g/L}$ ) on every incoming ARA oil lot for infant formula applications.                              | Immediate | QA / Incoming Inspection  |
| P1   | Quarantine and test all existing ARA oil inventory; prioritise batches of Chinese origin or produced 2024–2025.                                                           | 30 days   | QA / Supply Chain         |
| P1   | Map every SKU exposure to CABIO Biotech ARA oil; brief customers and executive team on contagion risk.                                                                    | 30 days   | Supply Chain / Commercial |
| P1   | Update ARA oil specifications: cereulide limit, EU Implementing Regulation compliance, supplier cereulide certificate, <i>ces</i> -gene environmental monitoring clauses. | 30 days   | Procurement / QA          |

| PRI. | ACTION                                                                                                                                                                 | TIMELINE | OWNER                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|
| P1   | Conduct unannounced supplier audits focused on fermentation facility hygiene, <i>B. cereus</i> environmental monitoring, and cereulide-suppressive process conditions. | 60 days  | QA / Supplier Management |
| P2   | Build strategic inventory buffer of 3–6 months qualified ARA oil.                                                                                                      | 90 days  | Supply Chain             |
| P2   | Deploy fraud-detection battery: GC-FID fatty-acid profiling + SIRA + DNA barcoding ( <i>M. alpina</i> ).                                                               | 90 days  | QA / Laboratory          |

### Next (3–12 months)

| PRI. | ACTION                                                                                                                                 | TIMELINE                 | OWNER                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| P2   | Qualify 2+ alternative ARA oil suppliers (prioritise DSM-Firmenich Kingstree; evaluate non-China options).                             | 12–24 months (start now) | Procurement / Regulatory |
| P3   | Pilot blockchain / digital-twin traceability for ARA oil (1–2 suppliers) capturing lot-level fermentation, testing and transport data. | 6–9 months               | IT / Supply Chain        |
| P3   | Crisis simulation — tabletop exercise of ARA-driven contamination recall spanning QA, legal, comms and commercial.                     | 6 months                 | Quality / Legal / Comms  |
| P3   | Evaluate algal / precision-fermentation alternative ARA sources.                                                                       | 12 months                | R&D / Regulatory         |
| P3   | Review product-recall and contingent business-interruption insurance coverage against ingredient-origin exclusions.                    | 6 months                 | Risk / Finance / Legal   |

### Later (12+ months)

| PRI. | ACTION                                                                                                 | TIMELINE     | OWNER                  |
|------|--------------------------------------------------------------------------------------------------------|--------------|------------------------|
| P3   | Engage EFSA / FDA / Codex on harmonised cereulide limits for fermentation-derived nutritional oils.    | 12–18 months | Regulatory Affairs     |
| P3   | Assess strategic options for vertically integrated or equity-stake ARA sourcing (EU or North America). | 24–36 months | Strategy / Procurement |

## 13. Glossary

| ABBREVIATION | EXPANSION                                                                   |
|--------------|-----------------------------------------------------------------------------|
| ARA          | Arachidonic acid (C20:4 n-6).                                               |
| ARfD         | Acute Reference Dose (health-based guidance value for short-term exposure). |
| AV           | Anisidine Value (secondary oxidation indicator).                            |
| BCP          | Border Control Post (EU point of entry for controlled imports).             |
| BOM          | Bill of Materials.                                                          |
| CAPA         | Corrective and Preventive Action.                                           |

| ABBREVIATION    | EXPANSION                                                                 |
|-----------------|---------------------------------------------------------------------------|
| CCP             | Critical Control Point (HACCP).                                           |
| <i>ces</i> gene | Gene cluster encoding cereulide biosynthesis in emetic <i>B. cereus</i> . |
| CFSA            | China National Center for Food Safety Risk Assessment.                    |
| DHA             | Docosahexaenoic acid (C22:6 n-3).                                         |
| ECDC            | European Centre for Disease Prevention and Control.                       |
| EFSA            | European Food Safety Authority.                                           |
| EMP             | Environmental Monitoring Programme.                                       |
| FSMA            | U.S. Food Safety Modernization Act.                                       |
| GRAS            | Generally Recognised As Safe (U.S. FDA).                                  |
| GRN             | GRAS Notice (FDA numbered filing).                                        |
| LCPUFA          | Long-Chain Polyunsaturated Fatty Acid.                                    |
| LC-MS/MS        | Liquid Chromatography – Tandem Mass Spectrometry.                         |
| MRL             | Maximum Residue Level.                                                    |
| NHC             | China National Health Commission.                                         |
| OSI             | Oxidation Stability Index.                                                |
| PV              | Peroxide Value (primary oxidation indicator).                             |
| RASFF           | Rapid Alert System for Food and Feed (EU).                                |
| SIP/CIP         | Sterilisation In Place / Cleaning In Place.                               |
| SIRA            | Site-Specific Isotope Ratio Analysis (authentication).                    |
| Totox           | Total oxidation value = 2×PV + AV.                                        |
| WGS             | Whole-Genome Sequencing.                                                  |

## 14. References & Sources

### Regulatory & Official Guidance

- WHO, Disease Outbreak News — DON596, "Multi-country foodborne event — cereulide in infant formula," 25 February 2026. [who.int/emergencies/disease-outbreak-news/item/2026-DON596](https://www.who.int/emergencies/disease-outbreak-news/item/2026-DON596)
- EFSA, "Scientific opinion on cereulide in infant and follow-on formula — ARfD and derived maximum levels," February 2026. [efsa.europa.eu](https://efsa.europa.eu)
- EFSA, "Precautionary global recall of infant nutrition products — Q&A," January 2026.
- ECDC, "Multi-country foodborne event caused by cereulide in infant formula — rapid risk assessment," February 2026.
- European Commission DG SANTE, newsroom 924278, "Infant formula reinforced import controls," February 2026. [ec.europa.eu/newsroom/sante](https://ec.europa.eu/newsroom/sante)